

Code-Layout, Readability and Reusability

Date	10 JULY 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

This document evaluates the quality of the **Emotion Detection & Learning Support Engine** source code in terms of structure, readability, maintainability, and reusability. The project follows a modular architecture where preprocessing, emotion prediction, recommendation generation, and user interface are implemented as separate components. Proper naming conventions, organized folder structure, reusable functions, and consistent coding practices improve code quality and simplify future maintenance and feature enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation and formatting throughout the codebase.	YES	Follows Python PEP-8 style..
2	Proper File Structure	Files and folders are logically organize	YES	Separate files for preprocessing, models, API integration, and U
3	Meaningful Variable Names	Variables reflect their purpose clearly	YES	Uses descriptive names for readability.
4	Function / Method Names	Functions are descriptively named	YES	Improves maintainability.
5	Code Comments	Important logic is documented using comments where required.	PARTIAL	Comments added for complex functions.
6	Modular Design	Application divided into reusable modules.	YES	Independent components for preprocessing, prediction, logging, and recommendations
7	No Redundant Code	Duplicate or unused code is removed	YES	Common functionality reused through helper modules.

8	Error Handling	Exceptions and errors are handled gracefully	YES	Includes model validation and API exception handling
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Text Preprocessing Module	PYTHON	Emotion prediction pipeline	HIGH

2	BERT Classifier	Hugging Face Transformers	Emotion classification	HIGH
3	BiLSTM Classifier	TensorFlow / Keras	Emotion Prediction	HIGH
4	Gemini Helper Module	Google Gemini AP	Personalized learning recommendations	HIGH
5	CSV Logger	Python	Prediction history storage	MEDIUM

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular architecture.
Readability	5	Clear naming conventions and consistent formatting
Reusability	5	Independent reusable modules for major functionalities.
Documentation / Comments	4	Adequate comments with scope for additional documentation.
Overall Score	5	Clean, scalable, and maintainable implementation following modular software design principles.